HANG (HARVEY) YU, Ph.D.

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PROFESSIONAL SUMMARY

Senior Machine Learning Engineer and Technical Lead with 10+ years of experience spanning Apple, GIC, and Walmart Labs, specializing in mission-critical decisioning systems, adversarial risk and fraud mitigation, and LLM/RAG engineering. Proven track record defending high-throughput production systems against sophisticated abuse, implementing guardrails for generative AI deployments, and driving resilient distributed architectures. Backed by rigorous computational and probabilistic modeling foundations with a Ph.D. in Computational Biophysics and an M.S. in Statistics from UIUC. Targeting the AI Security Bootcamp (AISB) 2027 to deepen expertise in frontier AI threat modeling, adversarial robustness, injection/jailbreak defense, and AI control.

AREAS OF EXPERTISE

- AI & Frontier Risk Interests: AI stack threat modeling, LLM jailbreaks & prompt/RAG injection defense, guardrail systems, adversarial robustness, formal verification.
- Adversarial & Risk ML: Fraud prediction & mitigation, anomaly detection, synthetic content & gibberish generation detection, account and warranty abuse mitigation.
- Production Engineering: High-throughput real-time decisioning, resilient distributed systems, Python, Clojure, PyTorch, Spark, Snowflake, SQL.
- Mathematical Foundations: Stochastic modeling, Bayesian inference, time-series analysis, high-performance scientific computing (HPC).

PROFESSIONAL EXPERIENCE

APPLE | Singapore Tech Lead / Machine Learning Engineer | March 2020 – Present

Adversarial Risk & AI Abuse Mitigation:

- Designed and maintained predictive machine learning pipelines and text-mining models to detect order fraud, malicious abuse patterns, and high-risk transactional vectors.
- Developed ML and LLM-based detection strategies against adversarial behavior, including automated gibberish-email generation and complex warranty-fraud rings.
- Collaborated closely with the Austin fraud engineering team on the Bamboo platform deployment; led technical investigations and debugging on adversarial edge cases for Technical Fraud Analysts (TFAs).

LLM Safety, Guardrails & Engineering:

- Architected and deployed an internal LLM-powered Retrieval-Augmented Generation (RAG) self-service platform for enterprise operational workflows, reducing manual support burden.
- Designed structured input/output guardrails, validation probes, and access constraints to mitigate prompt injection, context leakage, and unconstrained action execution.

Mission-Critical Real-Time Decisioning & Systems Resilience:

- End-to-end owner of APAC rollouts for real-time fulfillment decisioning engines (Creola, Delphi/PSLA), successfully achieving 100% regional coverage across APAC region.
- Defended decision thresholds and operational risk-reward trade-offs to executive steering committees using ROC-curve analysis and operating-point sensitivity modeling.
- Re-architected core replenishment algorithms to scale calculation throughput by 1,000x, enabling multi-node regional expansion from a single distribution center to all of APAC.
- Safeguarded 24/7 system reliability and zero-downtime operations during multi-infrastructure migrations (Snowflake, Erzhou) and severe logistical disruptions; pre-emptively identified a critical 30-minute cutoff mismatch that prevented widespread order delivery failures.

Technical Leadership & Mentoring:

- Mentored and onboarded MLEs and Technical Fraud Analysts in defensive system design, Python best practices, and production triage playbooks.
- Spearheaded cross-regional architectural alignment across PAC, EMEIA, and AMR teams.

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GIC | Singapore Data Scientist / Assistant Vice President (AVP) | February 2017 – March 2020

- Extracted alpha and risk signals from large-scale alternative datasets using NLP (Spark, NLTK) and deep learning frameworks (TensorFlow).
- Engineered geospatial computer vision pipelines for economic activity analysis and population distribution estimation using satellite image classification.
- Formulated predictive time-series models to analyze macro trends and asset volatility.

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WALMART LABS | Sunnyvale, CA Senior Software Engineer / Data Scientist | July 2015 – February 2017

- Developed core components of a distributed semantic search engine and query-parsing architecture using Solr, Hadoop, Hive, and Key-Value storage backends.
- Built high-throughput text classification and product attribute extraction models, driving a 4% lift in add-to-cart conversion (projected $5B annual revenue impact).

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EXXONMOBIL RESEARCH | Houston, TX Computational Scientist Intern | May 2014 – August 2014

- Designed and optimized high-performance machine learning algorithms on massive geological datasets, improving subsurface prediction accuracy by 30%.

EDUCATION

University of Illinois Urbana-Champaign (UIUC) | Urbana, IL

- Ph.D. in Computational Biophysics (2009 – 2015) Focus: Supercomputing molecular dynamics simulations, stochastic trajectory analysis, and high-dimensional clustering. Eugene Rabinowitch Graduate Fellowship recipient.
- M.S. in Statistics (2012 – 2015) Focus: Machine learning theory, stochastic processes, and large-scale predictive modeling.

Nanyang Technological University (NTU) | Singapore

- B.S. (First Class Honours) in Biology / Computing (2003 – 2007) Full Undergraduate Scholarship (Ministry of Education); Dean's List (2005–2007); President Research Scholarship.

INVITED TALKS & PUBLICATIONS

- Invited Speaker: "MCAA AsiaConnect: Innovations in Computer Science and AI" (Nov 2024)
- Author of peer-reviewed computational biophysics papers in Biophysical Journal and ACS publications covering molecular dynamics simulations and macromolecular interactions.
- Editorial Board Member: Universal Journal of Physics and Application (2015 – Present)

TECHNICAL SKILLS

- Languages & Core Tools: Python, PyTorch, Clojure, R, SQL, C/C++, Spark, Snowflake, Git
- AI & Security Focus: Threat Modeling, LLM/RAG Guardrails, Anomaly & Fraud Detection, Adversarial Robustness, Real-Time Decision Systems, High-Performance Computing